

ema_cross - Strategy Walkthrough

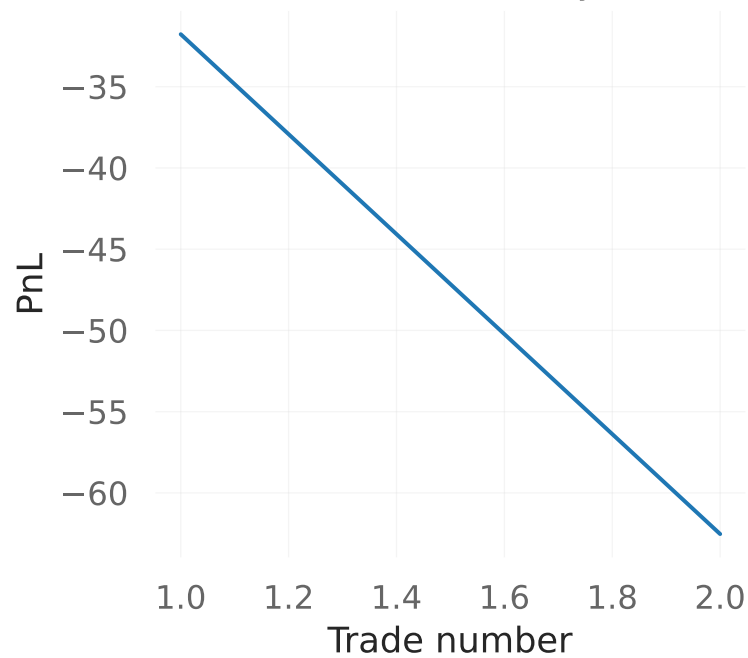
Strategy name `ema_cross` Files reviewed - `src/strategies/ema_cross/strategy.py` - `src/strategies/ema_cross/backtest_ema_cross_v2.py` - `src/strategies/ema_cross/config.py` - `src/strategies/ema_cross/runner.py` Step-by-step execution flow 1) Backtest fetches OHLCV and `ema_fast` and `ema_slow` columns. 2) Optional ATR and ADX series are prepared for filters and ATR stops. 3) BacktesterV2 is initialized with fee/slippage, direction, risk, pyramiding, and size controls. 4) For each bar from `ema_slow + 1` onward: `bt.on_bar` runs first for intrabar stop/take-profit handling. 5) Strategy `check_signal` runs on historical slice and computes `ema_fast`, `ema_slow`, `ema_trend` (default `trend_period=200`). 6) Entry or exit signal is generated from EMA cross log trend filter. 7) For entry signals, ATR non-NaN is required if ATR mode is enabled; `ADX >= adx_threshold` is required if ADX filter is configured. 8) Signal executes via `bt.on_signal` with optional `at` and `stop_loss_pct`. 9) Backtest exports stats, equity curve image, report, and trades CSV; optional trade plots are generated. 10) `runner.py` provides a live loop variant that waits for a new candle and prints LONG/SHORT/EXIT transitions. Parameters and defaults - Strategy cross inputs: `ema_fast` (required), `ema_slow` (required), `trend_period` is fixed by strategy default at 200. - ATR and ADX filters: `atr_period=None`, `atr_sl_mult=None`, `atr_tp_mult=None`, `adx_period=None`, `adx_threshold=None`. - Execution and cost controls: `initial_balance=1000.0`, `commission_pct=0.001`, `slippage_pct=0.01`. - Position controls: `position_mode="all_in"`, `trade_size=100.0`, `position_pct=None`, `pyramiding=1`. - Direction and risk controls: `allow_short=True`, `stop_loss_pct=0.02`, `take_profit_pct=None`. - Live config defaults (`config.py`): `EXCHANGE="binance"`, `SYMBOL="BTC/USDT"`, `TIMEFRAME="15m"`, `EMA_FAST=20`, `EMA_SLOW=50`, `POSITION_SIZE=100`. Indicators/calculations - EMA fast and EMA slow crossover. - EMA trend filter (EMA200 by default in `strategy.py`). - Optional ATR and optional ADX filter in backtest runner. Entry conditions - Long when `ema_fast` crosses above `ema_slow` and `close > ema_trend`. - Short when `ema_fast` crosses below `ema_slow` and `close < ema_trend`. Exit conditions - If upward cross occurs while `current_side` is SHORT and `stop_loss` is hit, strategy returns EXIT (short exit). - If downward cross occurs while `current_side` is LONG and `stop_loss` is hit, strategy returns EXIT (long exit). - No explicit non-cross exit in strategy; additional exits may come from BacktesterV2 stop/take-profit. Risk and position management - Percent `stop_loss_pct` and optional `take_profit_pct` are passed to BacktesterV2. - Optional ATR-based stops/take-profits when `atr_period` and `atr_sl_mult` are provided. - No custom trailing-stop logic in this strategy backtest. - Pyramiding supported by Backtest (default 1). - Position sizing by BacktesterV2 mode and optional `position_pct`. Outputs (trades, reports) - `clean_stats` from `bt.stats()`. - Equity curve PNG under `static/backtests/ema_cross/<run_id>/.` - Trades CSV with run metadata. - QuantStats report when `plot_stats=True`. Optional `plot_trades_by_date` and `plot_trades` outputs. Known ambiguities or inconsistencies - Loop starts at `ema_slow + 1`, but `check_signal` enforces `max(ema_slow, 200) + 1` bars; with slow `ema_slow` causes a long no-signal warmup period. - `backtest_ema_cross_v2` precomputes `ema_fast/ema_slow` then `check_signal` recomputes them on each slice. - Live runner uses `config.POSITION_SIZE` but does not pass that value anywhere in current signal handling.

ema_cross - Trade summary

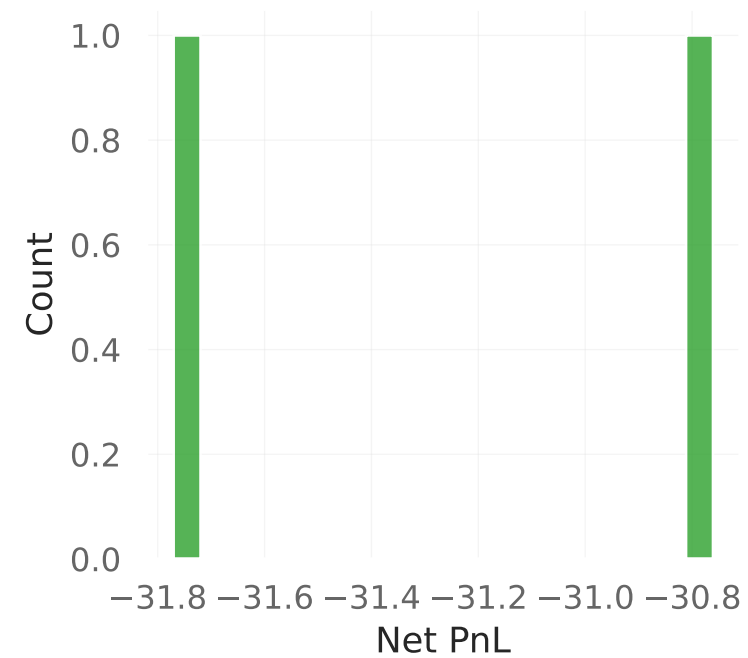
Trades: 2 Total Net PnL: -62.53 Average Net PnL per trade: -31.27 Win rate: 0.00% First trigger: 2017-01-01 15:00:00
crossed ABOVE EMA50 + EMA200 filter Last trigger: 2017-01-01 15:00:00 Source CSV: docs/strategy_walkthroughs/csv_baselines/ema_cross.csv

ema_cross - Trade analytics

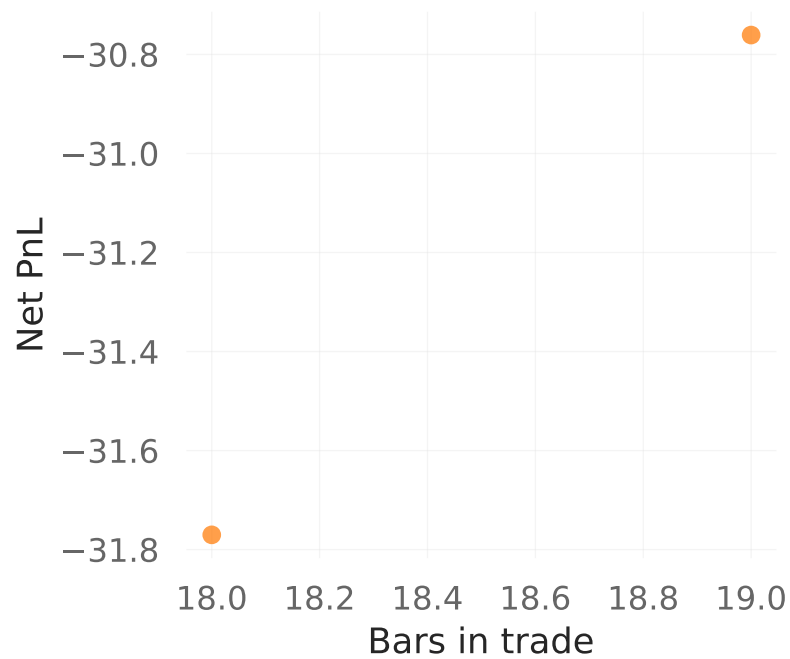
Cumulative Net PnL by trade



Net PnL distribution



Duration vs Net PnL



EMA20 crossed ABOVE EMA50 + EMA200 filter

Top entry triggers

